

WEEK-11

Task-1: Develop a multithreaded counter application using POSIX threads (pthread create, pthread_join). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently.

```
the_boss@DESKTOP-I8C8AJG: × + ▾
GNU nano 8.7.1 race.c *
#include <stdio.h>
#include <pthread.h>
|
#define THREADS 4
#define COUNT 100000

int counter = 0;

void *increment(void *arg) {
    for (int i = 0; i < COUNT; i++) {
        counter++;
    }
    return NULL;
}

int main() {
    pthread_t threads[THREADS];

    for (int i = 0; i < THREADS; i++)
        pthread_create(&threads[i], NULL, increment, NULL);

    for (int i = 0; i < THREADS; i++)
        pthread_join(threads[i], NULL);

    printf("Final counter = %d\n", counter);
    printf("Expected counter = %d\n", THREADS * COUNT);

    return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

Output:

```
the_boss@DESKTOP-I8C8AJG:~$ cd practical
the_boss@DESKTOP-I8C8AJG:~/practical$ nano race.c
the_boss@DESKTOP-I8C8AJG:~/practical$ gcc race.c -o race -pthread
the_boss@DESKTOP-I8C8AJG:~/practical$ ./race
Final counter = 227368
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ ./race
Final counter = 400000
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ ./race
Final counter = 297551
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ ./race
Final counter = 228607
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ |
```

Report: A multithreaded counter application was implemented using POSIX threads. Multiple threads concurrently modified a shared counter without synchronization. Different executions produced inconsistent results due to race conditions. This demonstrated the need for synchronization when accessing shared data.

Task-2: Modify the previous program using mutex locks (pthread mutex lock, pthread mutex_unlock) and compare the outputs with the race- condition version. Analyze the impact of synchronization.

```
the_boss@DESKTOP-I8C8AJG: × + v
GNU nano 8.7.1 mutex.c *
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define COUNT 100000

int counter = 0;

pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;

void *increment(void *arg) {
    for (int i = 0; i < COUNT; i++) {

        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main() {
    pthread_t threads[THREADS];

    for (int i = 0; i < THREADS; i++)
        pthread_create(&threads[i], NULL, increment, NULL);

    for (int i = 0; i < THREADS; i++)
        pthread_join(threads[i], NULL);

    printf("Final counter = %d\n", counter);
    printf("Expected counter = %d\n", THREADS * COUNT);

    pthread_mutex_destroy(&lock);
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T E
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J J
```

Output:

```
the_boss@DESKTOP-I8C8AJG:~/practical$ gcc mutex.c -o mutex -pthread
the_boss@DESKTOP-I8C8AJG:~/practical$ ./mutex
Final counter = 400000
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ ./mutex
Final counter = 400000
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ ./mutex
Final counter = 400000
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ ./mutex
Final counter = 400000
Expected counter = 400000
the_boss@DESKTOP-I8C8AJG:~/practical$ |
```

Report: The counter program was modified to use a POSIX mutex for synchronization. `pthread_mutex_lock()` and `pthread_mutex_unlock()` protected the shared counter from simultaneous access.

Unlike the race-condition version, the synchronized program consistently produced the expected result.

Synchronization improves correctness and reliability, although mutex operations introduce some execution overhead.